

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q24-2374

Commonwealth Chesapeake 138 kV  
March 29, 2025

### ***1. Introduction***

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2374. The Interconnection Customer proposes a 85.0 MW solar photovoltaic facility interconnecting at the Commonwealth Chesapeake 138 kV substation in the Virginia area. The request was received on January 20, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 487 higher-queued active requests and their assigned network upgrades.

### ***2. Summary of Results***

No thermal overloads attributable to this project were identified. No network upgrades are required. Estimated network upgrade cost: \$0

### ***3. Network Impacts***

#### ***3.1 Generator Deliverability and Single Contingency Analysis***

No overloads identified.

#### ***3.2 Multiple Facility Contingency Analysis***

No additional overloads were identified for multiple facility contingencies.

#### ***3.3 Short Circuit and Stability***

Short circuit and stability analyses identified no additional network upgrades.

### ***4. Required Network Upgrades and Cost Estimates***

None. Total network upgrade cost allocated to this project: \$0.

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.